

Geometry  
Unit 12  
Lesson 8 Practice

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

For questions 1 and 2, consider the equation of the circle represented by  
 $(x - 8)^2 + (y - 10)^2 = 12^2$ .

1. Rewrite the equation in general form.

$$\begin{aligned}(x - 8)^2 + (y - 10)^2 &= 12^2 \\ x^2 - 16x + 64 + y^2 - 20y + 100 &= 144 \\ x^2 + y^2 - 16x - 20y + 20 &= 0\end{aligned}$$

2. Give the key features of the circle.

| Center              | Diameter            | Radius |
|---------------------|---------------------|--------|
| (8, 10)             | 24                  | 12     |
| Domain              | Range               |        |
| $-4 \leq x \leq 20$ | $-2 \leq y \leq 22$ |        |

For questions 3 and 4, consider the equation of the circle represented by  
 $(x + 10)^2 + (y - 3)^2 = 1^2$ .

3. Rewrite the equation in general form.

$$\begin{aligned}(x + 10)^2 + (y - 3)^2 &= 1^2 \\ x^2 + 20x + 100 + y^2 - 6y + 9 &= 1 \\ x^2 + y^2 + 20x - 6y + 108 &= 0\end{aligned}$$

4. Give the key features of the circle.

| Center               | Diameter          | Radius |
|----------------------|-------------------|--------|
| (-10, 3)             | 2                 | 1      |
| Domain               | Range             |        |
| $-11 \leq x \leq -9$ | $2 \leq y \leq 4$ |        |

For questions 5 and 6, consider the equation of the circle represented by  
 $x^2 + y^2 - 18y = -65$ .

5. Rewrite the equation in center-radius form.

$$\begin{aligned}x^2 + y^2 - 18y &= -65 \\ x^2 + y^2 - 18y + 81 &= -65 + 81 \\ x^2 + (y - 9)^2 &= 16\end{aligned}$$

6. Give the key features of the circle.

| Center             | Diameter           | Radius |
|--------------------|--------------------|--------|
| (0, 9)             | 8                  | 4      |
| Domain             | Range              |        |
| $-4 \leq x \leq 4$ | $5 \leq y \leq 13$ |        |

For questions 7 and 8, consider the equation of the circle represented by

$$x^2 + y^2 + 10x = -21.$$

7. Rewrite the equation in center-radius form.

$$\begin{aligned} x^2 + 10x + y^2 &= -21 \\ x^2 + 10x + 25 + y^2 &= -21 + 25 \\ (x + 5)^2 + y^2 &= 4 \end{aligned}$$

8. Give the key features of the circle.

| Center              | Diameter           | Radius |
|---------------------|--------------------|--------|
| $(-5, 0)$           | 4                  | 2      |
| Domain              | Range              |        |
| $-7 \leq x \leq -3$ | $-2 \leq y \leq 2$ |        |

For questions 9 and 10, consider the equation of the circle represented by

$$x^2 + y^2 - 2x + 10y = 23.$$

9. Rewrite the equation in center-radius form.

$$\begin{aligned} x^2 - 2x + y^2 + 10y &= 23 \\ x^2 - 2x + 1 + y^2 + 10y + 25 &= 23 + 1 + 25 \\ (x - 1)^2 + (y + 5)^2 &= 49 \end{aligned}$$

10. Give the key features of the circle.

| Center             | Diameter            | Radius |
|--------------------|---------------------|--------|
| $(1, -5)$          | 14                  | 7      |
| Domain             | Range               |        |
| $-6 \leq x \leq 8$ | $-12 \leq y \leq 2$ |        |